

Nahid Hasan Sagar

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PROFESSIONAL SUMMARY

M.S. candidate in Statistical Science at Indiana University Bloomington with experience in computational statistics, reproducible data analysis, and benchmarking research workflows. Applies R, Python, C++, MATLAB, and Bash to support quantitative research, and efficient analysis pipelines for real-world data.

EDUCATION

Indiana University, College of Arts and Sciences

Master of Science in Statistical Science

Thesis: Time-Series Analysis Using Bayesian Additive Regression Trees (BART)

Graduate Coursework: Introduction to Statistical Computing; Applied Statistical Computing; Fundamentals of Statistics I–II; Applied Linear Models I–II

Current Coursework: Statistical Consulting; Research in Statistics; Introduction to Analysis I

University of Dhaka

Bachelor of Science in Physics

Bloomington, IN

Expected May 2027

Dhaka, Bangladesh

August 2024

EXPERIENCE

Research Assistant, StatMIND Lab

Indiana University Bloomington

2025 – Present

Bloomington, IN

- Developed core functionality of Bayesian brain mapping algorithms developed by Dr. Amanda F. Mejia with the implementation of efficient C++ code with Rcpp and RcppArmadillo for high-dimensional neuroimaging data.
- Reproduced the exact results of the original R package implementation and validated the correctness of the new C++ code through extensive unit testing and benchmarking.
- Achieved a 4 to 5× speedup in core functional computations with proper implementation of C++ code inside R packages, enabling faster analysis of large-scale neuroimaging datasets.
- Build reproducible benchmarking workflows to compare runtime, scalability, and computational performance across implementations.

Teaching Assistant, Department of Statistics

Indiana University Bloomington

2025 – Present

Bloomington, IN

- Support undergraduate Statistics courses through classroom instruction, discussion sections, and individualized guidance.
- Evaluate approximately 200 assignments per grading cycle and provide detailed feedback on statistical reasoning, computation, and interpretation.
- Conduct weekly office hours and exam-review sessions to strengthen students' conceptual understanding and problem-solving skills.

SELECTED RESEARCH & PROJECTS

M.S. Thesis: Time-Series Analysis Using BART

R, Bayesian Modeling

- Conduct thesis research on the use of Bayesian Additive Regression Trees for time-series analysis under the supervision of Dr. Matthew T. Pratola.
- Apply statistical modeling and reproducible analysis workflows to evaluate predictive performance on time-series data.

Ordinary Least Squares from Scratch

Python, NumPy, Pandas, Matplotlib

- Develop an educational implementation of ordinary least squares regression from first principles, emphasizing matrix-based estimation and transparent statistical computation.
- Validate outputs with careful numerical checks and clear visualization of model fit and residual behavior.

TECHNICAL SKILLS

Statistical Methods: Bayesian inference, BART, MCMC, hierarchical modeling, time-series analysis, linear and logistic regression, mixed-effects models, bootstrap methods, experimental design, model selection, and predictive model evaluation

Research and Analysis: Benchmarking, reproducible analysis workflows, statistical reporting, visualization, and literature review

Programming: R, Python, C, C++, MATLAB, Bash, HTML, CSS, and JavaScript

Computing and Tools: Unix/Linux, numerical optimization, algorithm benchmarking, code profiling, Git, GitHub, LaTeX, RStudio, and VS Code

Languages: Bengali (native), English (fluent), Hindi (conversational), Urdu (conversational)

LEADERSHIP & HONORS

- Elected as Graduate vice chairperson of the Statistics Club at Indiana University Bloomington for the 2026–2027 academic year.
- Attended the 2026 Velocity Conference at UC Berkeley as an Indiana University representative for the science community.
- Collaborated on the development of a comprehensive online Linear Algebra curriculum during the COVID-19 pandemic.
- Co-organized the Australian Mathematics Competition in Bangladesh in 2014.